

MATH 4162/Math 6201
Numerical methods for differential equations
Fall 2025

Homework problem set #2

Show all necessary steps of working in the given space

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Upload the assignment solution no later than

23:59 on Oct 24, 2025

Full marks 50

1. [25pts]

(a) For the advection-diffusion equation

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}, \quad a > 0 \quad \nu > 0$$

write the numerical scheme using upwind for advection and center for diffusion.

(b) Show that the exact amplification factor

$$G = \frac{u(x, t + \Delta t)}{u(x, t)}$$

satisfies $|G| = 1$ for the Fourier mode

$$u(x, t) = e^{i(kx + \omega t)}.$$

(c) Using von Neumann method, show that amplification factor of the scheme is

$$G = 1 - (C + 2D)(1 - \cos \beta) - Ci \sin \beta$$

where

$$C = \frac{a\Delta t}{\Delta x}, \quad D = \frac{\nu\Delta t}{\Delta x^2}, \quad \beta = k\Delta x.$$

(d) Plot the stability curve $G(\beta)$ by taking $C + 2D = 1$ and $C = 1$ for $\beta \in [-\pi, \pi]$.

Based on this plot, if the maximum wavenumber associated with the Fourier mode is $k_{\max} = 1000$, what should be the value of Δx .

Choose $C + 2D = 1/2$ and $C = 1$. Justify if the scheme will be stable for this choice of the parameters.

2. [25pts]

- (a) Using Taylor series expansion, find the leading order error terms for the given scheme (in Q1).
(b) Find the dispersion relation of the modified equation:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \underbrace{\left[\nu + \frac{a\Delta x}{2} - \frac{a^2\Delta t}{2} \right]}_{\nu_{eff}} \frac{\partial^2 u}{\partial x^2} + \underbrace{\left[\frac{-a\Delta x^2}{6} + a\nu\Delta t \right]}_{\phi_{eff}} \frac{\partial^3 u}{\partial x^3}$$

(which is given for equation and scheme in Q1).

- (c) Find the decay rate $e^{-\text{Im}\omega(k)t}$ of the modified equation and real part of the phase speed.
(d) Set $\phi_{eff} = 0$, determine the CFL number for the scheme to be stable?
Plot the stability curve $G(\theta)$ under this condition.
Show that $|G(\theta)| = 1$ requires $\Delta x = (4\nu)/a$.
(e) Select $\phi_{eff} = 0$. Using dispersion relation, show that this choice can eliminate the dispersion error; however, show that the dissipation error of numerical scheme is given by the rate $e^{-(2/3)\nu k^2 t}$

(a) For the advection-diffusion equation

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}, \quad a > 0 \quad \nu > 0$$

write the numerical scheme using upwind for advection and center for diffusion.

Solz

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + a \frac{u_i^n - u_{i-1}^n}{\Delta x} = \nu \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$$

$$\Rightarrow u_i^{n+1} - u_i^n + a \Delta t \frac{u_i^n - u_{i-1}^n}{\Delta x} = \nu \Delta t \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$$

$$\Rightarrow u_i^{n+1} = u_i^n - \underbrace{a \frac{\Delta t}{\Delta x} (u_i^n - u_{i-1}^n)}_{\text{advection}} + \underbrace{\nu \frac{\Delta t}{\Delta x^2} (u_{i+1}^n - 2u_i^n + u_{i-1}^n)}_{\text{diffusion}}$$

For advection,

$$u_i^n - a \frac{\Delta t}{\Delta x} u_i^n + a \frac{\Delta t}{\Delta x} u_{i-1}^n$$

For diffusion,

$$\nu \frac{\Delta t}{\Delta x^2} u_{i+1}^n - 2\nu \frac{\Delta t}{\Delta x^2} u_i^n + \nu \frac{\Delta t}{\Delta x^2} u_{i-1}^n$$

$$u_i^{n+1} = \left(a \frac{\Delta t}{\Delta x} + v \frac{\Delta t}{\Delta x^2} \right) u_{i-1}^n$$

$$+ \left(1 - a \frac{\Delta t}{\Delta x} - 2v \frac{\Delta t}{\Delta x^2} \right) u_i^n$$

$$+ \frac{v \Delta t}{\Delta x^2} u_{i+1}^n$$

(b) Show that the exact amplification factor

$$G = \frac{u(x, t + \Delta t)}{u(x, t)}$$

satisfies $|G| = 1$ for the Fourier mode

$$u(x, t) = e^{i(kx + \omega t)}.$$

Soln

$$G = \frac{u(x, t + \Delta t)}{u(x, t)}$$

$$G = \frac{e^{i(kx + \omega t + \omega \Delta t)}}{e^{i(kx + \omega t)}}$$

$$G = \frac{e^{i(kx + \omega t)} \cdot e^{i\omega \Delta t}}{e^{i(kx + \omega t)}}$$

$$G = e^{i\omega \Delta t}$$

$$|G| = |e^{i\omega \Delta t}|$$

$$\text{if } |e^{i\theta}| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$$

$$\therefore |e^{i\omega \Delta t}| = 1$$

$$\therefore |G| = 1$$

(c) Using von Neumann method, show that amplification factor of the scheme is

$$G = 1 - (C + 2D)(1 - \cos \beta) - Ci \sin \beta$$

where

$$C = \frac{a\Delta t}{\Delta x}, \quad D = \frac{\nu\Delta t}{\Delta x^2}, \quad \beta = k\Delta x.$$

Soln

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}$$

$$u_j^n = G^n e^{ijk\Delta x}$$

$$e^{i(j+1)\beta} = e^{ij\beta} e^{i\beta} = e^{ij\beta} (\cos \beta + i \sin \beta)$$

$$e^{i(j-1)\beta} = e^{ij\beta} e^{-i\beta} = e^{ij\beta} (\cos \beta - i \sin \beta)$$

$$\Rightarrow \frac{u_j^{n+1} - u_j^n}{\Delta t} + a \frac{u_j^n - u_{j-1}^n}{\Delta x} = \nu \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2}$$

$$\begin{aligned} \Rightarrow & \frac{G^{n+1} e^{ij\beta} - G^n e^{ij\beta}}{\Delta t} + a \frac{G^n e^{ij\beta} - G^n e^{i(j-1)\beta}}{\Delta x} \\ & = \nu \frac{G^n e^{i(j+1)\beta} - 2G^n e^{ij\beta} + G^n e^{i(j-1)\beta}}{\Delta x^2} \end{aligned}$$

$$\Rightarrow \frac{G-1}{\Delta t} + a \frac{1-e^{-i\beta}}{\Delta x} = v \frac{e^{i\beta}-2+e^{-i\beta}}{\Delta x^2}$$

we know, $1-e^{-i\beta} = 1-\cos\beta + i\sin\beta$
 $e^{i\beta}+e^{-i\beta}-2 = -2(1-\cos\beta)$

$$\Rightarrow G-1 = -C(1-\cos\beta + i\sin\beta) - D \cdot 2(1-\cos\beta)$$

$$\Rightarrow G = 1 - C(1-\cos\beta) - Ci\sin\beta - 2D(1-\cos\beta)$$

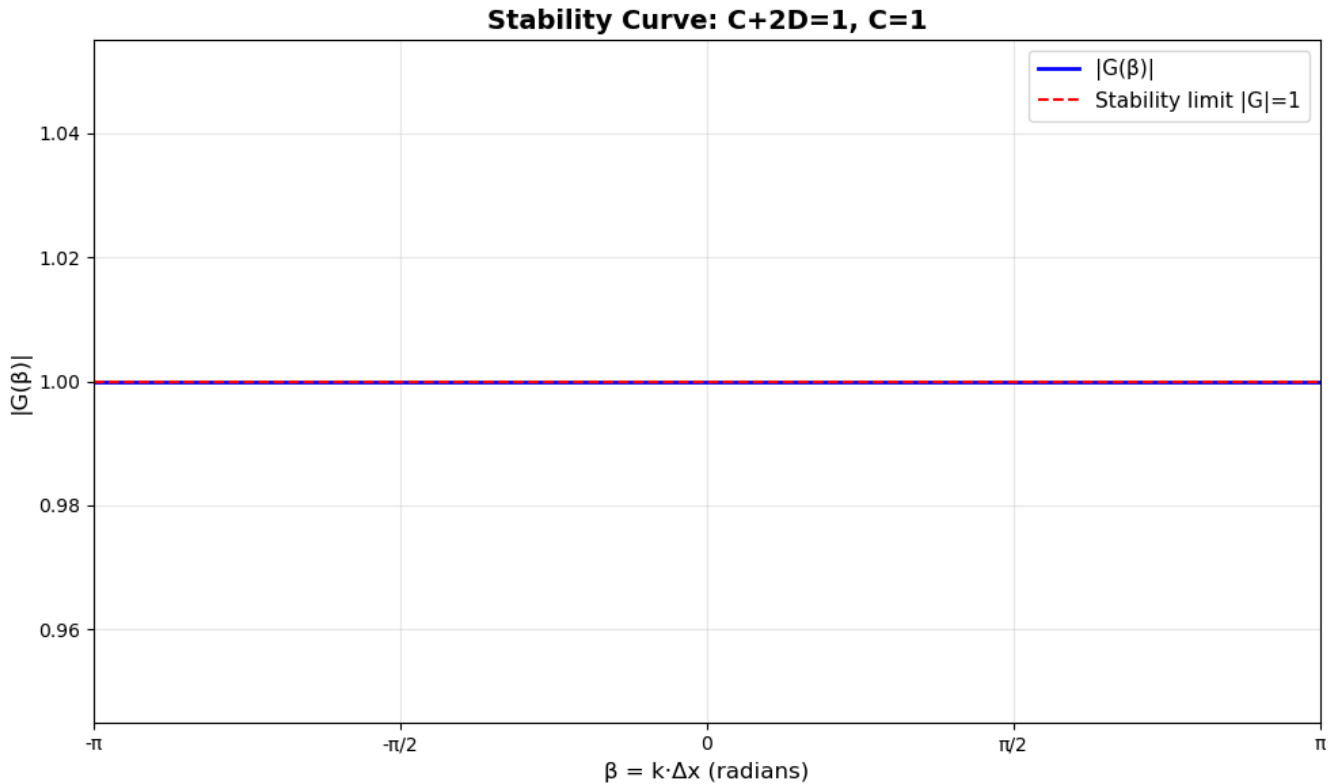
$$\Rightarrow G = 1 - (C+2D)(1-\cos\beta) - Ci\sin\beta$$

(d) Plot the stability curve $G(\beta)$ by taking $C + 2D = 1$ and $C = 1$ for $\beta \in [-\pi, \pi]$.

Based on this plot, if the maximum wavenumber associated with the Fourier mode is $k_{\max} = 1000$, what should be the value of Δx .

Choose $C + 2D = 1/2$ and $C = 1$. Justify if the scheme will be stable for this choice of the parameters.

Soln



$$G = \underbrace{1 - (C + 2D)(1 - \cos\beta)}_{\text{Re}} - \underbrace{C \sin\beta}_{\text{Im}}$$

For $C + 2D = 1, C = 1;$

$$\text{Re}(G) = 1 - (1 - \cos\beta) = \cos\beta$$

$$\text{Im}(G) = -\sin\beta$$

$$|G|^2 = \sin^2\beta + \cos^2\beta = 1$$

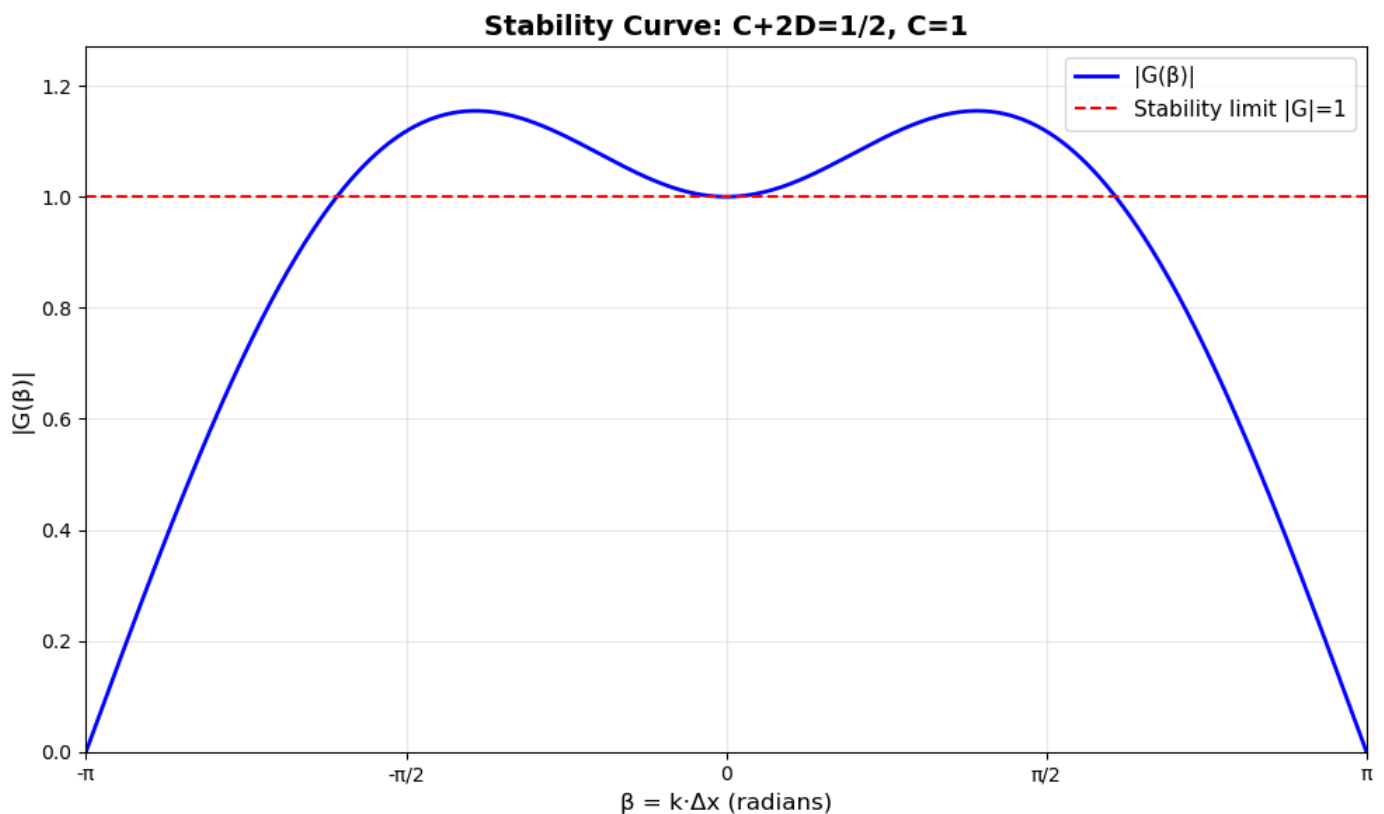
↑
Same as
1b.

for $k_{\max} = 1000;$

$$\beta_{\max} = \pi$$

$$\beta_{\max} = k_{\max} \cdot \Delta x$$

$$\Delta x = \frac{\beta_{\max}}{k_{\max}} = \frac{\pi}{1000} = 0.003142$$



$$\max |G(\beta)| = 1.1547 \quad \text{at} \quad \beta = -1.2296$$

Stability condition is $|G| \leq 1$

$\therefore$ This scheme will be unstable
for $C+2D = \frac{1}{2}$ & $C=1$.

2.

(a) Using Taylor series expansion, find the leading order error terms for the given scheme (in Q1).

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = v \frac{\partial^2 u}{\partial x^2} \quad a > 0, v > 0$$

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + a \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = v \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2}$$

Forward difference

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \left. \frac{\partial u}{\partial t} \right|_j^n + \frac{\Delta t}{2} \left. \frac{\partial^2 u}{\partial t^2} \right|_j^n + \frac{\Delta t^2}{6} \left. \frac{\partial^3 u}{\partial t^3} \right|_j^n + O(\Delta t^3)$$

Central difference

$$u_{j+1}^n = u_j^n + \Delta x \left. \frac{\partial u}{\partial x} \right|_j^n + \frac{\Delta x^2}{2} \left. \frac{\partial^2 u}{\partial x^2} \right|_j^n + \frac{\Delta x^3}{6} \left. \frac{\partial^3 u}{\partial x^3} \right|_j^n$$

$$+ \frac{\Delta x^4}{24} \left. \frac{\partial^4 u}{\partial x^4} \right|_j^n + O(\Delta x^5)$$

$$u_{j-1}^n = u_j^n - \Delta x \left. \frac{\partial u}{\partial x} \right|_j^n + \frac{\Delta x^2}{2} \left. \frac{\partial^2 u}{\partial x^2} \right|_j^n - \frac{\Delta x^3}{6} \left. \frac{\partial^3 u}{\partial x^3} \right|_j^n$$

$$+ \frac{\Delta x^4}{24} \left. \frac{\partial^4 u}{\partial x^4} \right|_j^n + O(\Delta x^5)$$

Advection term:

$$\Rightarrow \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = \left. \frac{\partial u}{\partial x} \right|_j^n + \frac{\Delta x^2}{6} \left. \frac{\partial^3 u}{\partial x^3} \right|_j^n + O(\Delta x^4)$$

Diffusion term:

$$\Rightarrow \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2} = \left. \frac{\partial^2 u}{\partial x^2} \right|_j^n + \frac{\Delta x^2}{12} \left. \frac{\partial^4 u}{\partial x^4} \right|_j^n + O(\Delta x^4)$$

Substitute into eqn.

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + a \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = v \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2}$$

$$\Rightarrow \left[\frac{\partial u}{\partial t} + \frac{\Delta t}{2} \frac{\partial^2 u}{\partial t^2} + O(\Delta t^2) \right]$$

$$+ a \left[\frac{\partial u}{\partial x} + \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + O(\Delta x^4) \right]$$

$$= v \left[\frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^2}{12} \frac{\partial^4 u}{\partial x^4} + O(\Delta x^4) \right]$$

$$\Rightarrow \frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} - v \frac{\partial^2 u}{\partial x^2} = - \frac{\Delta t}{2} \frac{\partial^2 u}{\partial t^2} - \frac{\Delta t}{2} \frac{\partial^2 u}{\partial t^2}$$

$$- a \frac{\Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} + v \frac{\Delta x^2}{12} \frac{\partial^4 u}{\partial x^4} + O(\Delta t^2, \Delta t^4)$$

$$\Rightarrow \text{Error} = \frac{\Delta t}{2} \frac{\partial^2 u}{\partial t^2} + \frac{a \Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{v \Delta x^2}{12} \frac{\partial^4 u}{\partial x^4} + O(\Delta t^2, \Delta t^4)$$

$$\frac{\partial u}{\partial t} = -a \frac{\partial u}{\partial x} + v \frac{\partial^2 u}{\partial x^2}$$

$$\Rightarrow \frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial t} \left(-a \frac{\partial u}{\partial x} + v \frac{\partial^2 u}{\partial x^2} \right)$$

$$\Rightarrow \frac{\partial^2 u}{\partial t^2} = -a \frac{\partial^2 u}{\partial t \partial x} + v \frac{\partial^3 u}{\partial t \partial x^2}$$

$$\Rightarrow \frac{\partial^2 u}{\partial x \partial t} = \frac{\partial}{\partial x} \left(-a \frac{\partial u}{\partial x} + v \frac{\partial^2 u}{\partial x^2} \right) = -a \frac{\partial^2 u}{\partial x^2} + v \frac{\partial^3 u}{\partial x^3}$$

$$\Rightarrow \frac{\partial^3 u}{\partial x^2 \partial t} = \frac{\partial}{\partial x^2} \left(-a \frac{\partial u}{\partial x} + v \frac{\partial^2 u}{\partial x^2} \right) = -a \frac{\partial^3 u}{\partial x^3} + v \frac{\partial^4 u}{\partial x^4}$$

$$\therefore \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} - 2av \frac{\partial^3 u}{\partial x^3} + v^2 \frac{\partial^4 u}{\partial x^4}$$

$$\Rightarrow \text{Error} = \frac{\Delta t}{2} \left(a^2 \frac{\partial^2 u}{\partial x^2} - 2av \frac{\partial^3 u}{\partial x^3} + v^2 \frac{\partial^4 u}{\partial x^4} \right) \\ + \frac{a \Delta x^2}{6} \frac{\partial^3 u}{\partial x^3} - \frac{v \Delta x^2}{12} \frac{\partial^4 u}{\partial x^4}$$

$$\therefore \text{Error} = \frac{a^2 \Delta t}{2} \frac{\partial^2 u}{\partial x^2} + \left(\frac{a \Delta x^2}{6} - av \Delta t \right) \frac{\partial^3 u}{\partial x^3} \\ + \left(\frac{v^2 \Delta t}{2} - \frac{v \Delta x^2}{12} \right) \frac{\partial^4 u}{\partial x^4} + \dots$$

(b) Find the dispersion relation of the modified equation:

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \underbrace{\left[\nu + \frac{a\Delta x}{2} - \frac{a^2\Delta t}{2} \right]}_{\nu_{eff}} \frac{\partial^2 u}{\partial x^2} + \underbrace{\left[\frac{-a\Delta x^2}{6} + a\nu\Delta t \right]}_{\phi_{eff}} \frac{\partial^3 u}{\partial x^3}$$

(which is given for equation and scheme in Q1).

Soln

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \left[\nu + \frac{a\Delta x}{2} - \frac{a^2\Delta t}{2} \right] \frac{\partial^2 u}{\partial x^2} + \left[\frac{-a\Delta x^2}{6} + a\nu\Delta t \right] \frac{\partial^3 u}{\partial x^3}$$

Assume $u(x,t) = \hat{u} e^{i(kx - \omega t)}$

$$\Rightarrow \frac{\partial u}{\partial t} = -i\omega \hat{u} e^{i(kx - \omega t)} = -i\omega u$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = (ik)^2 u = -k^2 u$$

$$\Rightarrow \frac{\partial^3 u}{\partial x^3} = (ik)^3 u = -ik^3 u$$

$$\therefore -i\omega u + a(iku) = v_{\text{eff}}(-k^2 u) + \phi_{\text{eff}}(-ik^3 u)$$

$$\Rightarrow -i\omega + aik = v_{\text{eff}}k^2 - i\phi_{\text{eff}}k^3$$

$$\Rightarrow -i\omega = -aik - v_{\text{eff}}k^2 - i\phi_{\text{eff}}k^3$$

$$\Rightarrow \omega = ak + \frac{v_{\text{eff}}k^2}{i} + \phi_{\text{eff}}k^3$$

$$\text{we know } \frac{1}{i} = -i$$

$$\Rightarrow \omega = ak - iv_{\text{eff}}k^2 + \phi_{\text{eff}}k^3$$

Dispersion Relation

$$\Rightarrow \omega(k) = ak + \phi_{\text{eff}}k^3 - iv_{\text{eff}}k^2$$

$$\Rightarrow \omega(k) = ak + \left[\frac{-a\Delta x^3}{6} + av\Delta t \right] k^3 - i \left[v + \frac{a\Delta x}{2} - \frac{a^2\Delta t}{2} \right] k^2$$

(c) Find the decay rate $e^{-\text{Im}\omega(k)t}$ of the modified equation and real part of the phase speed.

Soln

$$\text{Re}(\omega) = ak + \phi_{\text{eff}} k^3$$

$$\text{Im}(\omega) = -\nu_{\text{eff}} k^2$$

$$u = \hat{u} e^{i(kx - \omega t)} = \hat{u} e^{i(kx - \text{Re}[\omega]t - i\text{Im}[\omega]t)}$$

$$u(x, t) = \hat{u} e^{i(kx - \text{Re}[\omega]t)} e^{\text{Im}[\omega]t}$$

$$|u| = |\hat{u}| \cdot e^{\text{Im}[\omega]t}$$

$$|u| = |\hat{u}| \cdot e^{-\nu_{\text{eff}} k^2 t}$$

Decay rate

$$C_{\text{phase}} = \frac{\text{Re}(\omega)}{k} = \frac{ak + \phi_{\text{eff}} k^3}{k} = a + \phi_{\text{eff}} k^2$$

Phase speed

$$\Rightarrow C_{\text{phase}} = a + \left[\frac{-a \Delta x^2}{6} + a\nu \Delta t \right] k^2$$

- (d) Set $\phi_{eff} = 0$, determine the CFL number for the scheme to be stable?
 Plot the stability curve $G(\theta)$ under this condition.
 Show that $|G(\theta)| = 1$ requires $\Delta x = (4\nu)/a$.

Soln

If $\phi_{eff} = 0$,

$$\phi_{eff} = -\frac{a \Delta x^2}{6} + a v \Delta t = 0$$

$$\Rightarrow a v \Delta t = \frac{a \Delta x^2}{6}$$

$$\Rightarrow v \Delta t = \frac{\Delta x^2}{6}$$

$$\Rightarrow D = \frac{v \Delta t}{\Delta x^2} = \frac{1}{6}$$

Von Neumann;

$$G = 1 - (C + 2D)(1 - \cos \beta) - C i \sin \beta$$

$$\Rightarrow G = 1 - (C + 2(\frac{1}{6}))(1 - \cos \beta) - C i \sin \beta$$

We know,

$$|G|^2 = \text{Re}(G)^2 + \text{Im}(G)^2$$

$$\text{Re}(G) = 1 - \left(c + \frac{1}{3}\right)(1 - \cos\beta)$$

$$\text{Im}(G) = -C \sin\theta$$

$$\begin{aligned} \therefore |G|^2 &= \left[1 - \left(c + \frac{1}{3}\right)(1 - \cos\beta)\right]^2 \\ &\quad + C^2 \sin^2\beta \end{aligned}$$

$$\text{for stability, } |G|^2 = 1$$

$$\Rightarrow \left[1 - \left(c + \frac{1}{3}\right)(1 - \cos\beta)\right]^2 + C^2 \sin^2\beta = 1$$

$$\begin{aligned} \Rightarrow 1 - 2\left(c + \frac{1}{3}\right)(1 - \cos\beta) + \left(c + \frac{1}{3}\right)^2 (1 - \cos\beta)^2 \\ + C^2 \sin^2\beta = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow -2\left(c + \frac{1}{3}\right)(1 - \cos\beta) + \left(c + \frac{1}{3}\right)^2 (1 - \cos\beta)^2 \\ + C^2 \sin^2\beta = 0 \end{aligned}$$

$$\Rightarrow -2\left(C + \frac{1}{3}\right)(1 - \cos\beta) + \left(C + \frac{1}{3}\right)^2(1 - \cos\beta)^2 + C^2(1 - \cos\beta)(1 + \cos\beta) = 0$$

$$\Rightarrow 1 - \cos\beta \left[-2\left(C + \frac{1}{3}\right) + \left(C + \frac{1}{3}\right)^2(1 + \cos\beta) + C^2(1 + \cos\beta) \right] = 0$$

Assuming $\beta \neq 0$,

$$\Rightarrow -2\left(C + \frac{1}{3}\right) + \left(C + \frac{1}{3}\right)^2(1 - \cos\beta) + C^2(1 + \cos\beta) = 0$$

At critical point, $\beta = \pi$, $(1 - \cos\pi) = 2$
 $(1 + \cos\pi) = 0$

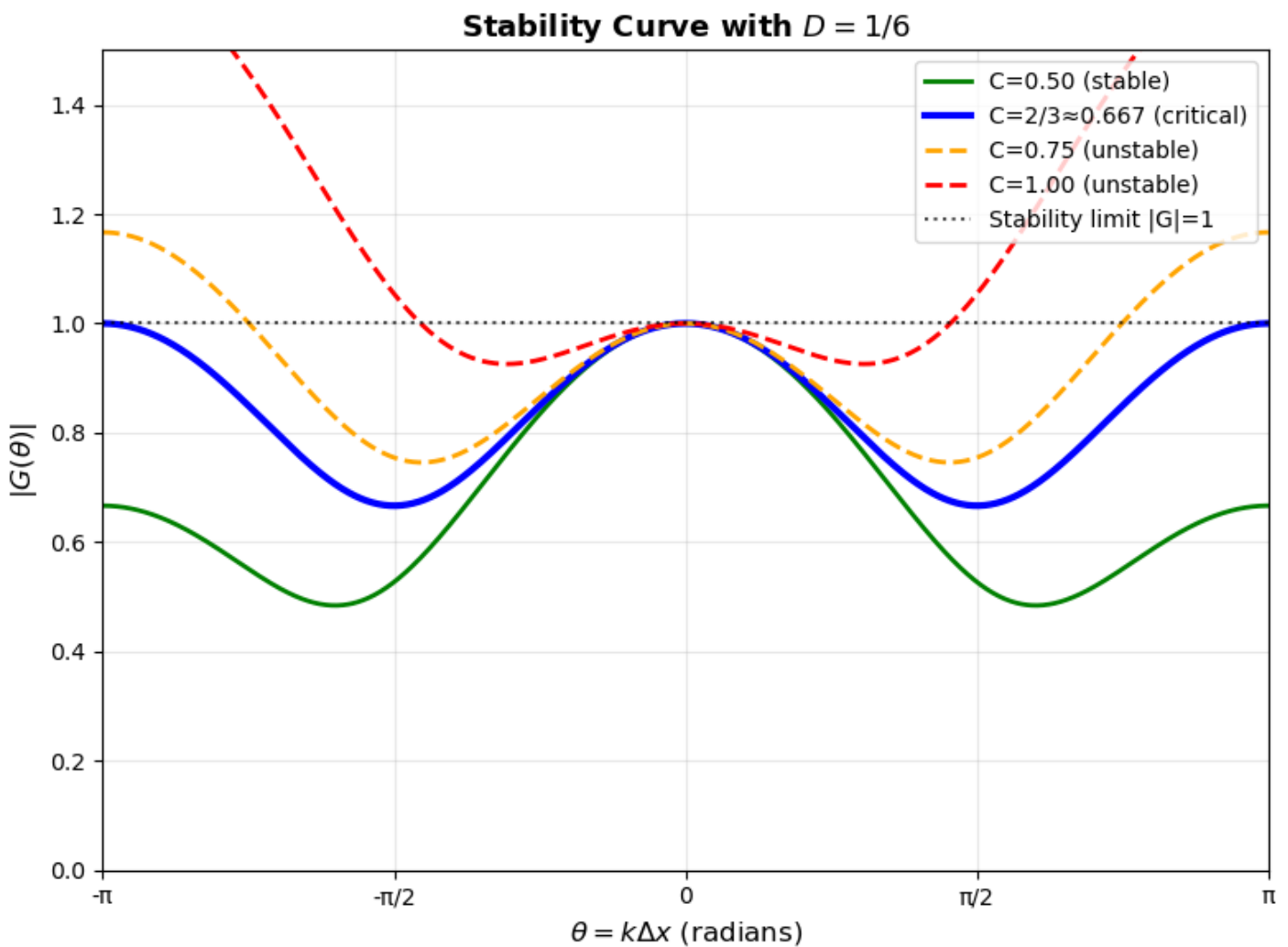
$$\Rightarrow -2\left(C + \frac{1}{3}\right) + 2\left(C + \frac{1}{3}\right)^2 = 0$$

$$\Rightarrow -2\left(C + \frac{1}{3}\right) \left[1 - \left(C + \frac{1}{3}\right) \right] = 0$$

$$\Rightarrow C + \frac{1}{3} = 1$$

$$\Rightarrow C = \frac{2}{3} = \frac{a \Delta t}{\Delta x}$$

$\therefore$ for $|G| \leq 1$, $C \leq \frac{2}{3}$
for all ϕ .



From $\phi_{\text{eff}} = 0$;

$$v \Delta t = \frac{\Delta x^2}{6} \quad \text{--- (1)}$$

From $C = \frac{2}{3}$

$$a \Delta t = \frac{2 \Delta x}{3} \quad \text{--- (2)}$$

$$\frac{(2)}{(1)} = \frac{a \Delta t}{v \Delta t} = \frac{a}{v} = \frac{\frac{2 \Delta x}{3}}{\frac{\Delta x^2}{6}} = \frac{2 \Delta x}{3} \cdot \frac{6}{\Delta x^2} = \frac{4}{\Delta x}$$

$$\frac{a}{v} = \frac{4}{\Delta x}$$

$$\Rightarrow \Delta x = \frac{4v}{a}$$

(e) Select $\phi_{eff} = 0$. Using dispersion relation, show that this choice can eliminate the dispersion error; however, show that the dissipation error of numerical scheme is given by the rate $e^{-(2/3)\nu k^2 t}$

$$\underline{Soln}$$

$$\omega = ak + \phi_{eff} k^3 - i\nu_{eff} k^2$$

$$\text{Say } \phi_{eff} = 0,$$

$$\omega = ak - i\nu_{eff} k^2$$

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}$$

$$u = \hat{u} e^{i(kx - \omega t)}$$

$$-i\omega + ak = \nu(-k^2)$$

$$\omega_{exact}(k) = ak - i\nu k^2$$

$$\omega_{num}(k) = ak - i\nu_{eff} k^2$$

$$\text{Re}[\omega_{exact}] = ak$$

$$C_{phase, exact} = \frac{ak}{k} = a$$

$$\text{Re}[w_{\text{num}}] = ak$$

$$C_{\text{phase, num}} = \frac{ak}{k} = a$$

$$C_{\text{phase, num}} = C_{\text{phase, exact}} = a$$

$$\text{Say } \phi_{\text{eff}} = 0,$$

$$\frac{-a \Delta x^2}{6} + av \Delta t = 0$$

$$\Rightarrow v \Delta t = \frac{\Delta x^2}{6}$$

$$\Rightarrow \beta = \frac{v \Delta t}{\Delta x^2} = \frac{1}{6}$$

$$\text{We know, } C = \frac{2}{3}$$

$$a \Delta t = \frac{2 \Delta x}{3}$$

$$v_{\text{eff}} = v + \frac{a \Delta x}{2} - \frac{a^2 \Delta t}{2}$$

$$v_{\text{eff}} = v + \frac{a \Delta x}{2} - \frac{a^2 \cdot \frac{2 \Delta x}{3a}}{2}$$

$$v_{\text{eff}} = v + \frac{a \Delta x}{2} - \frac{a \Delta x}{3}$$

$$v_{\text{eff}} = v + \frac{a \Delta x}{6}$$

we know $\Delta x = \frac{4v}{a}$

$$\Rightarrow v_{\text{eff}} = v + \frac{a \cdot \frac{4v}{a}}{6} = v + \frac{4v}{6} = \frac{5v}{3}$$

Now, $|u| = |\hat{u}| e^{\text{Im}[\omega]t}$

here, $\text{Im}[\omega] = -v_{\text{eff}} k^2 = -\frac{5v}{3} k^2$

$$\Rightarrow |u| = |\hat{u}| e^{-\frac{5v}{3} k^2 t}$$

$$|u_{\text{exact}}| = |\hat{u}| e^{-\nu k^2 t}$$

$$|u_{\text{num}}| = |\hat{u}| e^{-5/3 \nu k^2 t}$$

$$\text{Error} = e^{-5\nu/3 k^2 t} - e^{-\nu k^2 t} = e^{-2/3 \nu k^2 t}$$

↑
dissipation
error